Do Solinas-shape primes change the distribution of #E(F_p)?

Protocol: PROTOCOL.md, frozen in commit 588bdf5ac before the run. Instrument: examples/solinas_order_census.rs. Raw rows: runs/census.jsonl (one line per prime), summary runs/census.summary.txt, trace runs/census.log. Figure: figures/census_pvalues.svg. Conductor task T-4.

Result

Each prime had N = 4000 uniformly random curves $y^2 = x^3 + ax + b$, counted exactly. The sample was 29 Solinas-shape primes (2^n − 2^k ± 1, 16–22 bits, 6 ≤ k ≤ 17), 58 residue-matched random primes (same size, same p mod 480) and 58 generic random primes. Run time: 9 min 23 s on 4 threads (the protocol estimated about 20 min).

role	primes	F1 v_2(#E) rejects	F3 3 | #E rejects	F4 5 | #E rejects	F2 control rejects
Solinas-shape	29	0	0	0	29 / 29
residue-matched random	58	0	0	0	58 / 58
generic random	58	0	0	0	58 / 58

Bonferroni threshold for F1/F3/F4 over 435 tests at α = 0.01: $2.3 \cdot 10^{-5}$. The F2 control uses α / 145 = $6.9 \cdot 10^{-5}$; its largest p-value anywhere was $1.2 \cdot 10^{-6}$, and its median noncentrality was 76 on 4 degrees of freedom.

Verdict, in the protocol's wording. The instrument is valid (F2 rejected the wrong-class model for every prime). Solinas-shape primes are **within the matrix model for their own residue class at this resolution**, as are the residue-matched and generic primes. By §4 of the protocol this licenses "Solinas shape adds nothing beyond its residue class" for these three statistics at this size.

Descriptive, not part of the decision rule: the smallest target p-values were $6.0 \cdot 10^{-4}$ (F4), $2.5 \cdot 10^{-2}$ (F1) and $2.6 \cdot 10^{-2}$ (F3). With 435 tests the smallest expected p-value under the null is a few 10^{-3}, so none is unusual.

What the model says about the Solinas class (exact, no sampling)

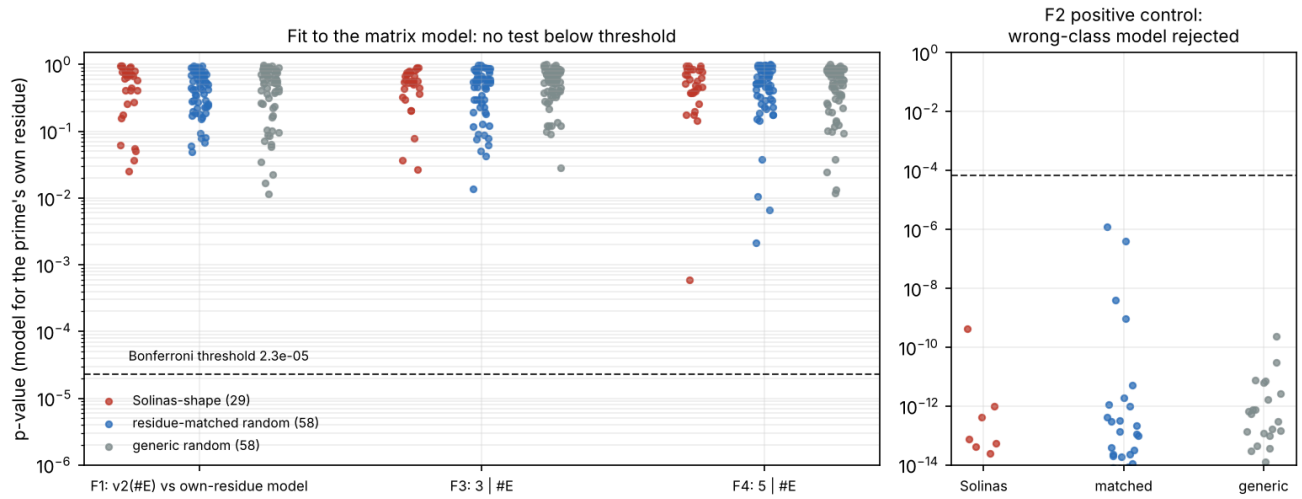
The enumeration gives a second result that does not depend on the sample. For all 29 targets, the total-variation distance between the prime's own-residue model of min(v_2(#E), 5) and the generic model (mean over primes with the same p mod 4) is 0 for the 21 targets with p ≡ 7 mod 8 and 0.0026 for the 8 with p ≡ 1 mod 8, whatever k is (6 to 17). So having p ≡ ±1 mod 2^k for large k changes the v_2(#E) distribution no more than p mod 8 already does.

What this does not show

- It does not test the claim at 224 bits, at a = −3, for structured curves (GLV, CM), for the odd part of #E beyond divisibility by 3 and 5, or for index-calculus relation yield. It says nothing about ECDLP cost.
- It cannot resolve a difference of the size the model itself predicts between the Solinas class and the generic class (≤ 0.003). That would need about 10^6 curves per prime. The result is "fits the model", not "differs from generic".
- A pattern the matrix model does not describe, for instance one that appears only at much larger p, would not be seen here.

Corrections made before the run

The first positive control (the generic model) had no power, because the generic model differs from the own-residue model by at most 0.003. It was replaced by the wrong-class model before any real data, and this is disclosed in PROTOCOL.md §6.



N = 4000 random curves per prime, 16-22 bit primes, seed 20261023. Source: runs/census.jsonl (PROTOCOL.md frozen in commit 588bdf5ac before the run).